

Autonomous Water Sampler for Contamination Research in Remote Locations



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Need for Water Sampling

- Dr. Mark Torres' lab studies cycling of water, carbon, and oxygen in surface environments
- Collect water samples daily to analyze metal concentrations surface environment safety
- Regular sampling tracks pollution levels for effective detection and management



Challenges of Sample Collection

- Remote locations make access difficult
- Need for consistent sampling methods
- Long period between sampling required (1 day)

Limitations of Current Samplers

- Exorbitant cost
- Inability to support:
 - 31 consecutive days of sampling
 - Extended periods of unattended operation

Objective

Create a portable, low cost, autonomous water sampling and storage device for geological researchers.

Design Parameters

Sampling Frequency
30 samples over a period of 1-2 months

Sample Size
>5ml

Weight, Volume & Dimensions
≤15kg, <90x50x45cm

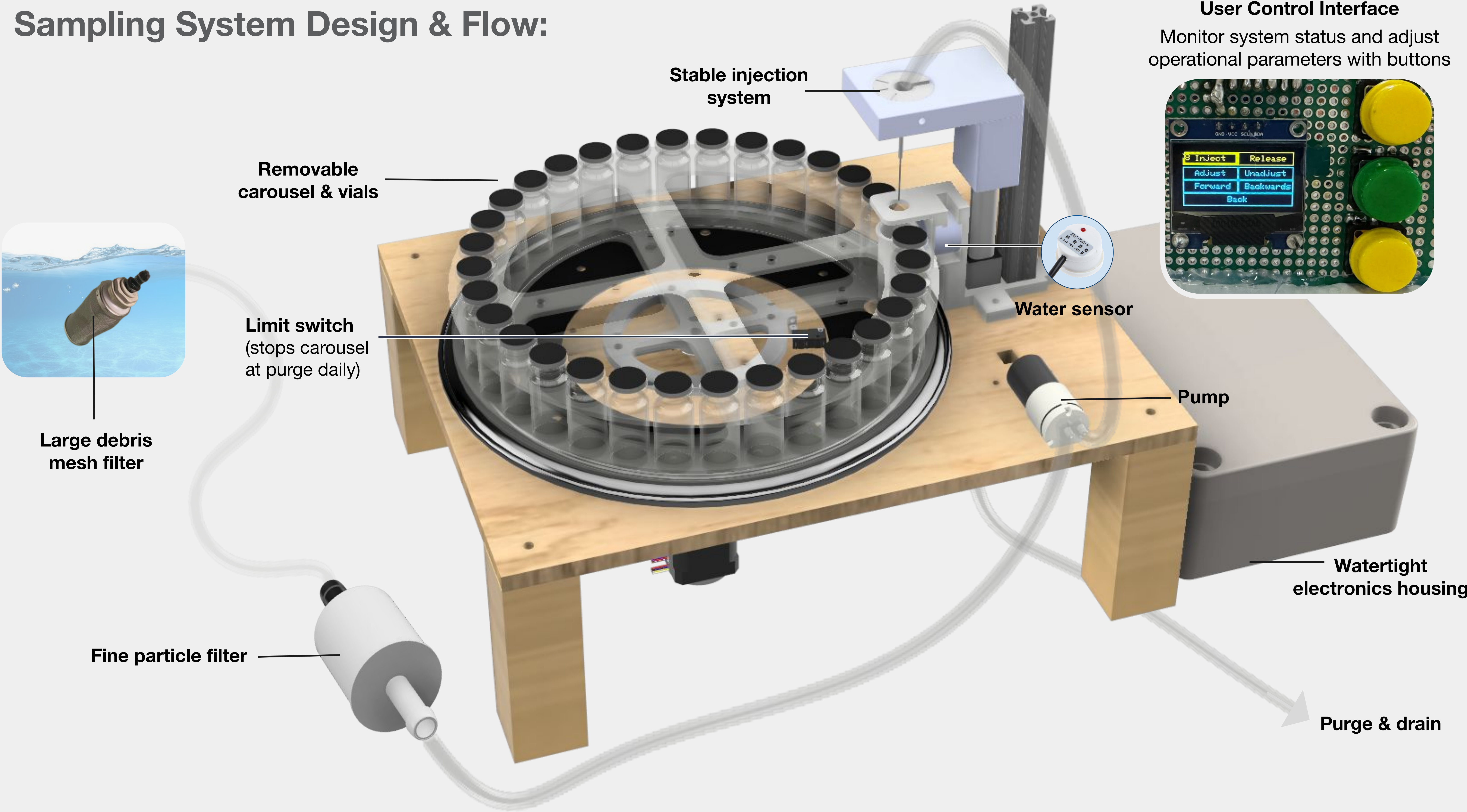
Particle Filtration
Removes all particles >0.45µm

Evaporation Mitigation
<1% evaporation within stored samples

Waterproofing
Standard Ingress Protection ≥IP65

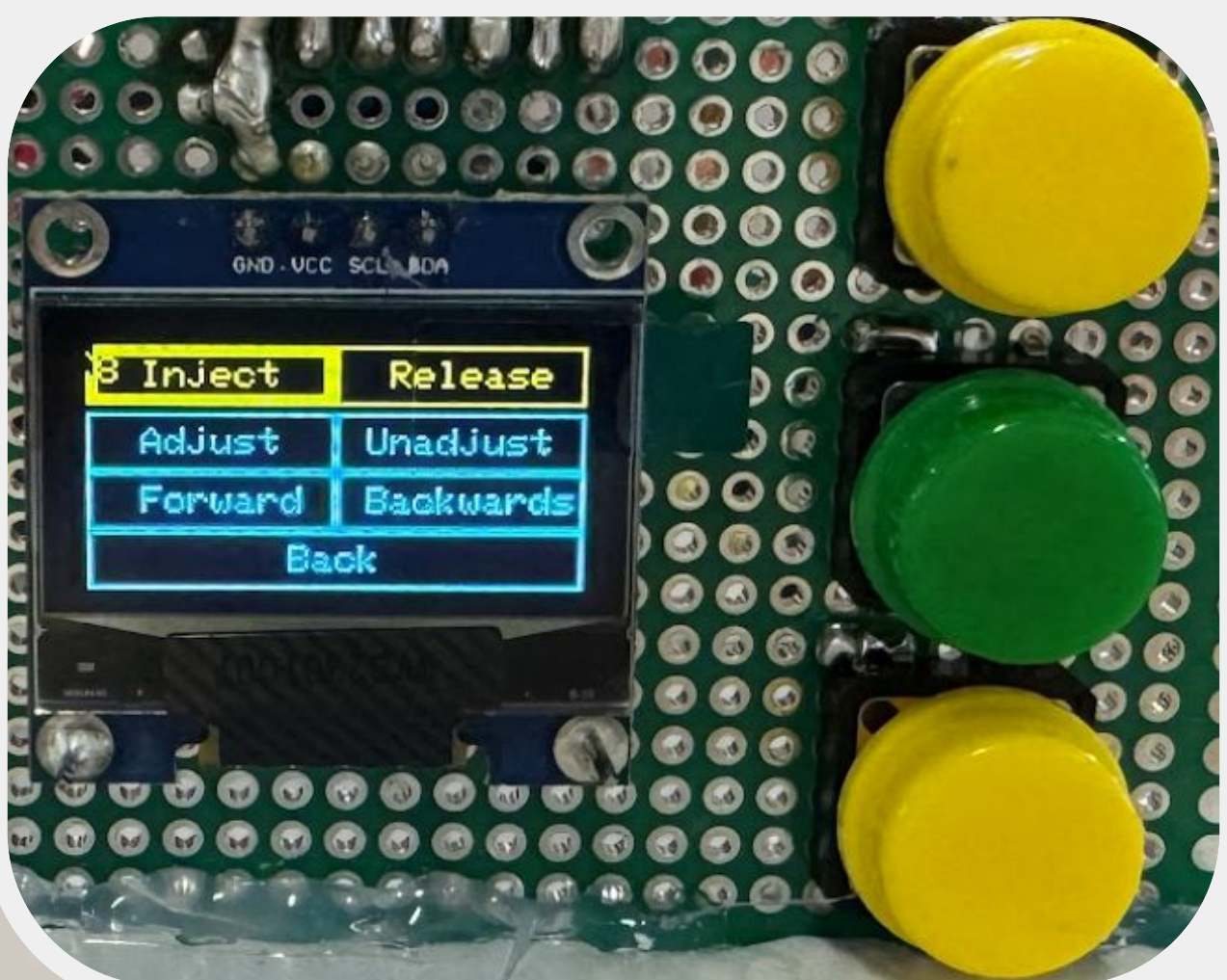
Sample Integrity
Leak-proof sample transport from water source to test tube

Sampling System Design & Flow:



User Control Interface

Monitor system status and adjust operational parameters with buttons



Timing Circuit

24-hour sleep cycle between samples

System Purge

New water pumped through to flush previous day's water

Carousel Movement

Stepper motor moves to today's vial

Vial Puncture

Linear actuator lowers needle to piercing self-healing lid

Sample Intake

Diaphragm pump pulls water through filters and fills vial

Full Vial Detection

Water level sensor indicates when vial is full, stopping pump and restarting sleep cycle

Testing

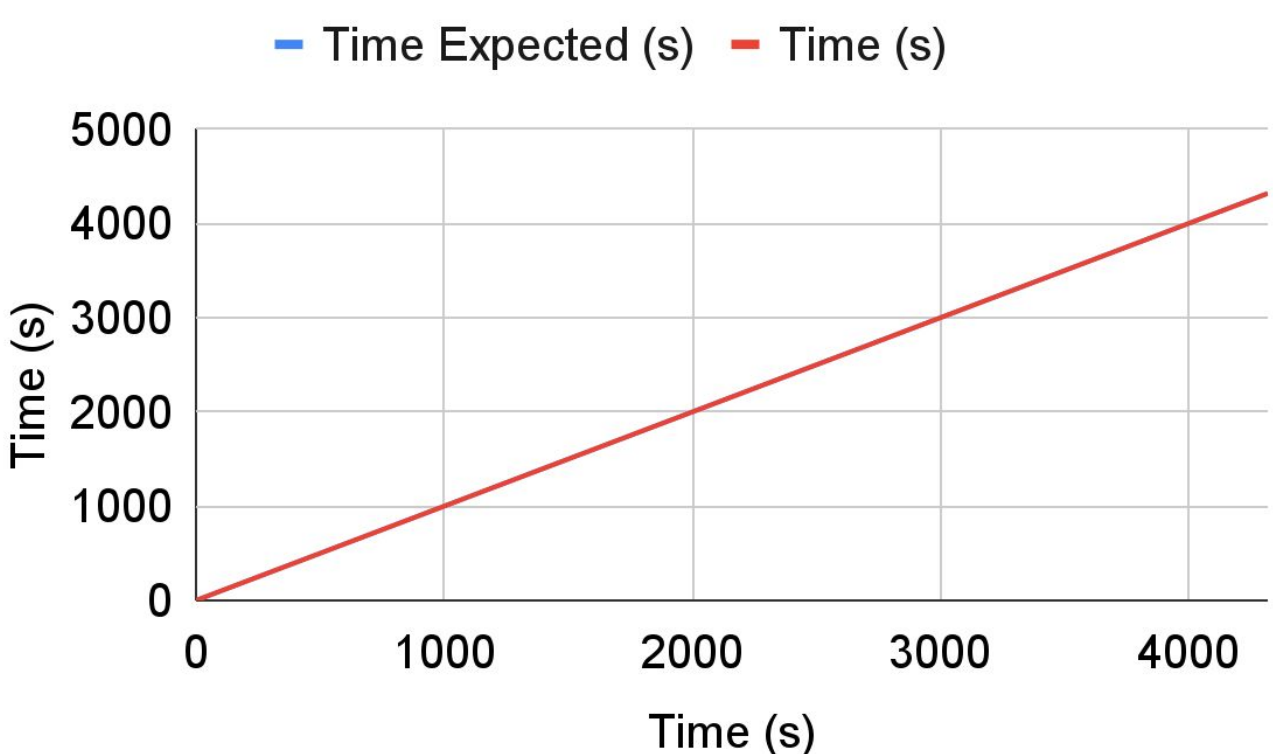
Pump

- Vertical suction height: > 6 ft
- Horizontal suction length: > 25 ft

Injection Accuracy

- Injections remained within the 3 mm central zone of the self-healing lids throughout full cycle

Clock



Impact

- ✓ Reduces need for unnecessary time and effort for researchers
- ✓ Improves precision of sampling
- ✓ Optimizes sampling for more samples and different frequencies
- ✓ Lowers costs

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